

Landmark CNN Architectures I

LeNet • AlexNet • VGG

Computer Vision
Week 4 – Lecture 7

Learning Objectives



Understand Evolution

Analyze why CNN architectures evolved over time, driven by data scale and hardware.



Compare Architectures

Identify the motivations, structural differences, and key innovations behind LeNet, AlexNet, and VGG.



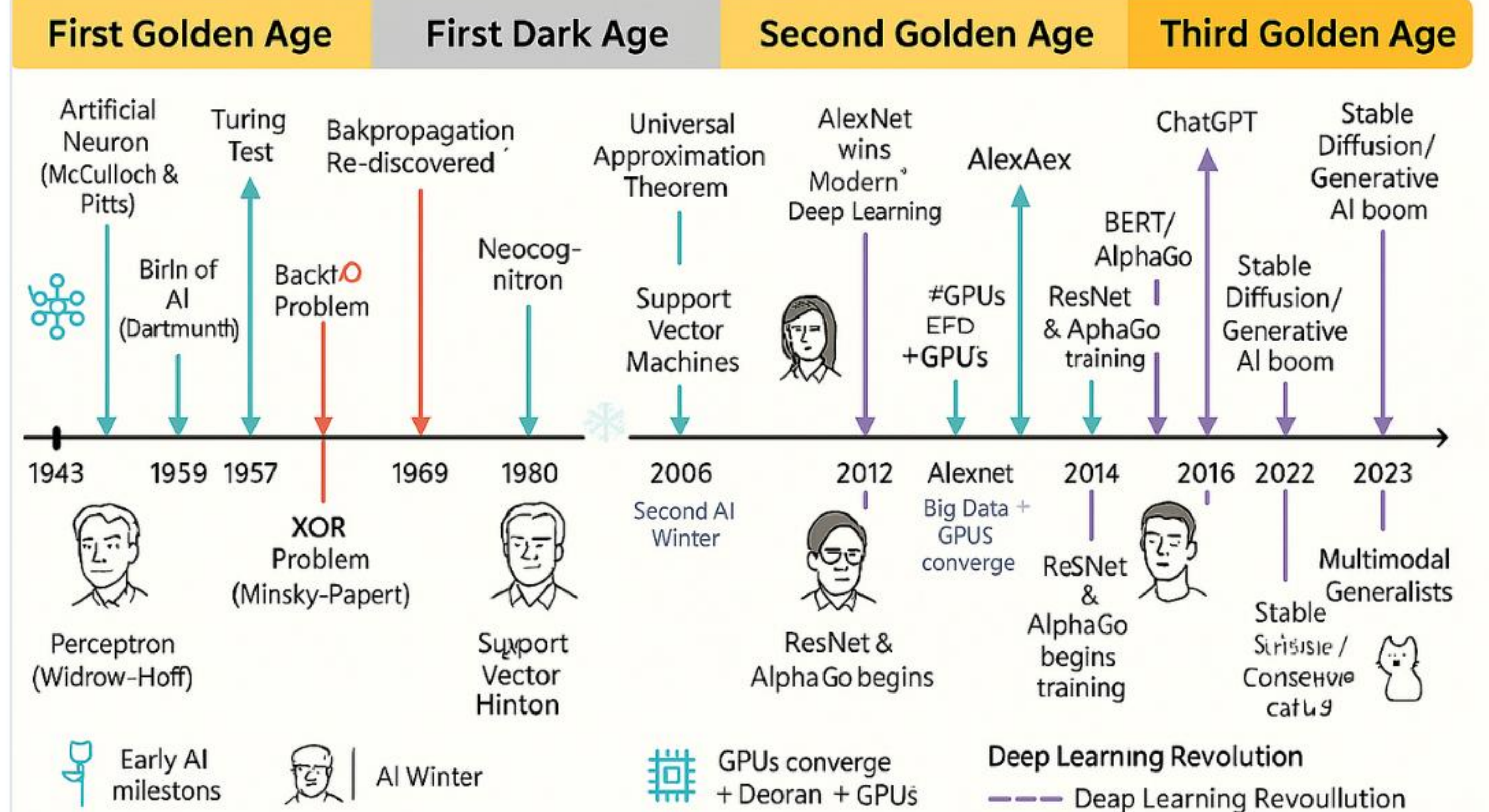
Select Appropriately

Learn to evaluate advantages and limitations to select the correct architecture for specific applications.

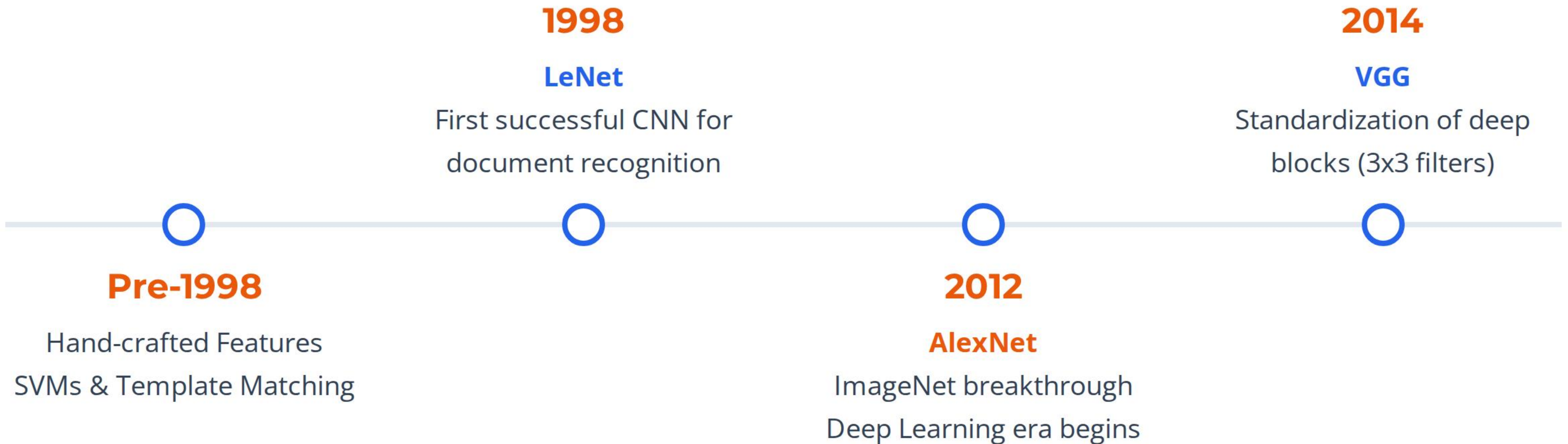
Why CNN Architectures Evolve

- 
Increasing Dataset Sizes: Transitioning from simple digits (MNIST) to complex real-world imagery (ImageNet).
- 
Computational Power: GPU advancements allowed the training of massive, highly parameterized networks.
- 
Need for Depth: Deeper networks are mathematically proven to learn more complex hierarchical features.
- 
Better Feature Extraction: Shift from shallow learning to highly abstract automatic feature learning.

A Brief History of AI & Deep Learning (1950 → 2025)



Historical Timeline



The Problem Before LeNet

-  **Hand-crafted Features:** Engineers had to manually design filters (e.g., Sobel, Gabor) for every specific problem.
-  **Template Matching:** Algorithms looked for exact pixel-by-pixel matches, which failed with varying lighting or angles.
-  **Poor Scalability:** Traditional machine learning models plateaued in accuracy regardless of how much data was provided.
-  **Need for Automation:** The field desperately needed a way for models to **learn optimal features automatically** from raw pixel data.

LeNet-5 Introduction

Origin & Developer

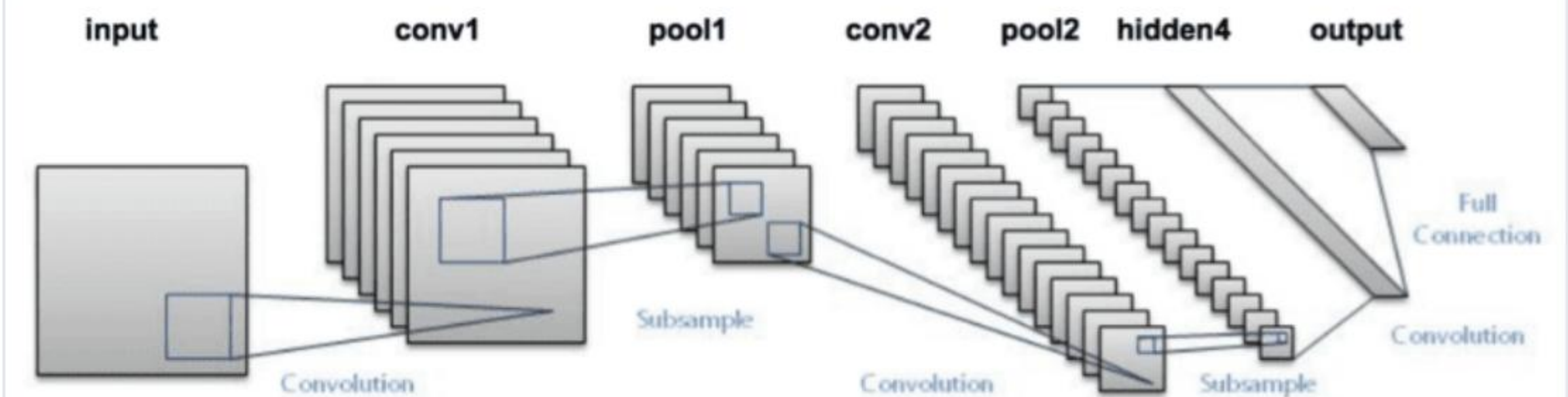
Developed by **Yann LeCun** and his team at Bell Labs in 1998. It is widely considered the foundational architecture that proved the viability of Convolutional Neural Networks.

Purpose & Dataset

Specifically designed for **digit recognition** tasks, primarily used to read zip codes on mail. It was trained on the famous **MNIST dataset** (32x32 grayscale images).

LeNet Architecture

- **Input:** 32x32 Grayscale Image
- **Convolution 1:** 6 filters (5x5)
- **Average Pooling 1:** 2x2 stride
- **Convolution 2:** 16 filters (5x5)
- **Average Pooling 2:** 2x2 stride
- **Fully Connected Layers:** 120 -> 84 -> 10 Output



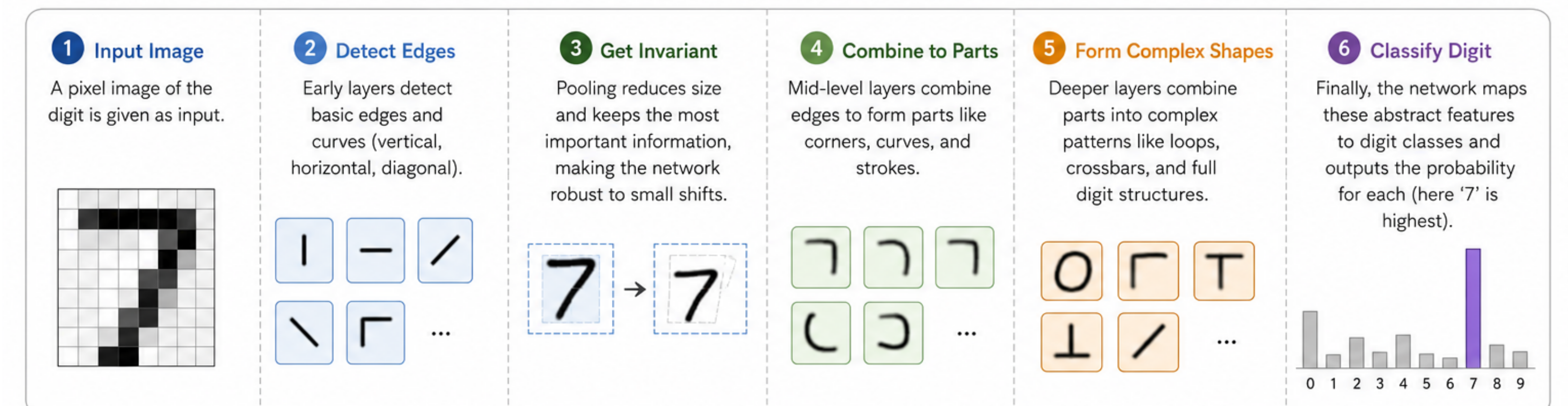
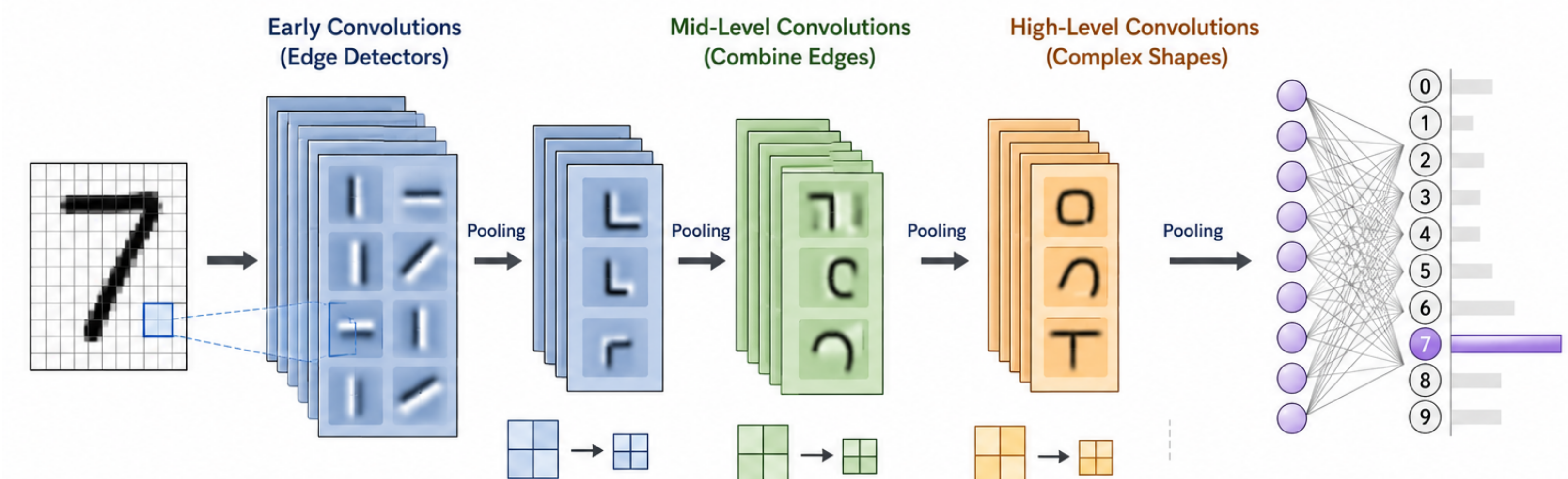
How LeNet Processes an Image

Early Layers: The first convolutions act as simple **edge and curve detectors**.

Pooling Layers: Reduce spatial dimensions, granting translation invariance (e.g., recognizing a '7' even if shifted).

Deep Layers: Feature maps combine edges into complex shapes like loops or crossbars.

Output: The network maps these abstract shapes to a specific digit class probability.



LeNet: Advantages & Limitations

✓ Advantages

- Extremely simple and easy to interpret.
- Very fast inference; runs easily on basic CPUs.
- Low computational requirements (~60,000 parameters).
- Highly effective for simple, low-variance datasets.

✗ Limitations

- Cannot handle high-resolution, complex color images.
- Shallow network limits feature abstraction capabilities.
- Use of Tanh/Sigmoid activations led to vanishing gradients if scaled up.

Transition to AlexNet

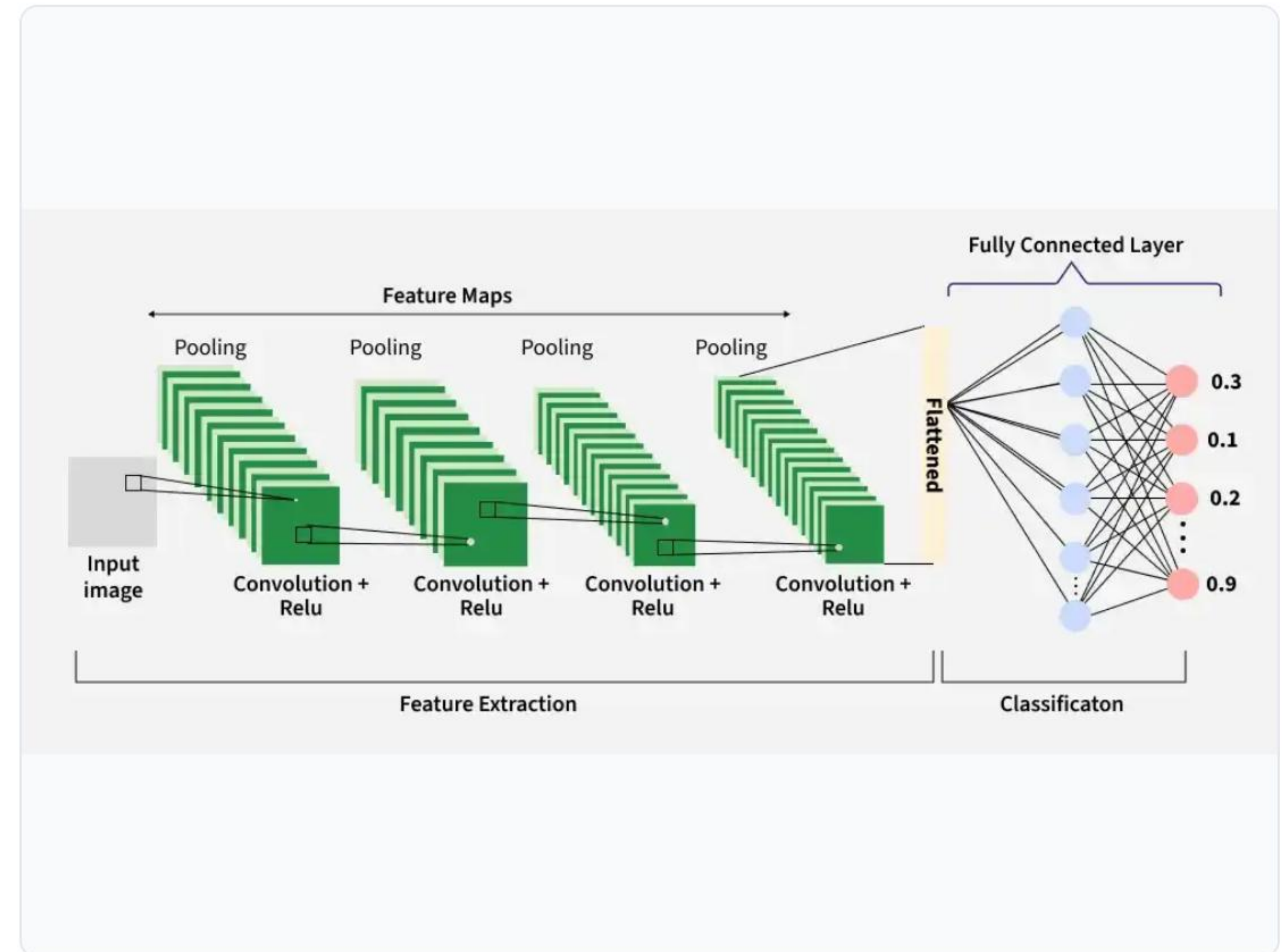
Overcoming the limitations of shallow networks through larger data, modern hardware, and architectural breakthroughs.

AlexNet Introduction

- 🏆 **2012 Breakthrough:** Won the ILSVRC (ImageNet Large Scale Visual Recognition Challenge) by a massive margin.
- 🖼️ **ImageNet Dataset:** Trained on over 1.2 million high-resolution color images across 1000 categories.
- 👥 **Creators:** Developed by Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton.
- ⚡ **The Catalyst:** This specific network is credited with triggering the modern Deep Learning revolution in Computer Vision.

AlexNet Architecture

- **Input:** 227x227x3 RGB Images
- **5 Convolutional Layers** (Sizes: 11x11, 5x5, 3x3)
- **Dual Pipeline:** Split across two GPUs for parallel training
- **Overlapping Max Pooling** for downsampling
- **3 Fully Connected Layers** (4096 neurons each)
- **Output:** 1000-way Softmax classifier



Key Innovations of AlexNet



ReLU Activation

Replaced Tanh with **Rectified Linear Units (ReLU)**. This massively accelerated training and solved the vanishing gradient problem.



Dropout

Introduced **Dropout** regularization in the FC layers to prevent severe overfitting on 60 million parameters.



Data Augmentation

Used image translations, horizontal reflections, and patch extractions to artificially expand the dataset and improve generalization.

Comparison: AlexNet vs LeNet

Metric	LeNet-5 (1998)	AlexNet (2012)
Dataset Target	MNIST (32x32 Grayscale)	ImageNet (227x227 Color)
Network Depth	7 layers (2 Conv, 3 FC)	8 layers (5 Conv, 3 FC)
Parameters	~60 Thousand	~60 Million
Activation Function	Tanh / Sigmoid	ReLU
Hardware Used	Standard CPU	Parallel GPUs (GTX 580)



Transition to VGG

The need for a simpler, uniform, yet vastly deeper architecture to capture extreme visual complexities.

VGG Network Introduction

Origin

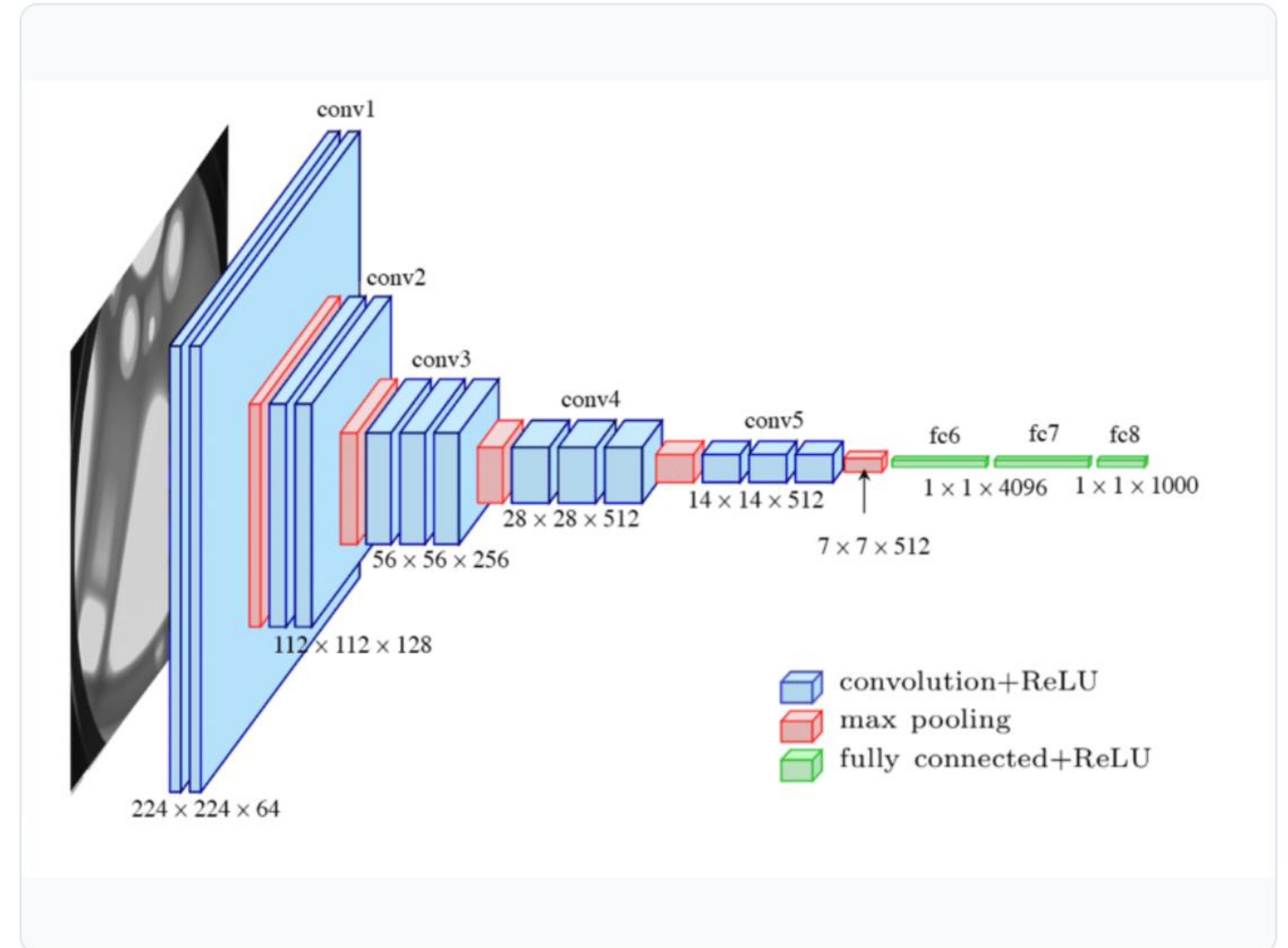
Developed by the **Visual Geometry Group (VGG)** at the University of Oxford in 2014. Secured top places in the 2014 ILSVRC competition.

The Core Philosophy

Replace large receptive fields (like AlexNet's 11x11) with **repeated, sequential 3x3 convolution filters** to increase depth while controlling parameter explosions.

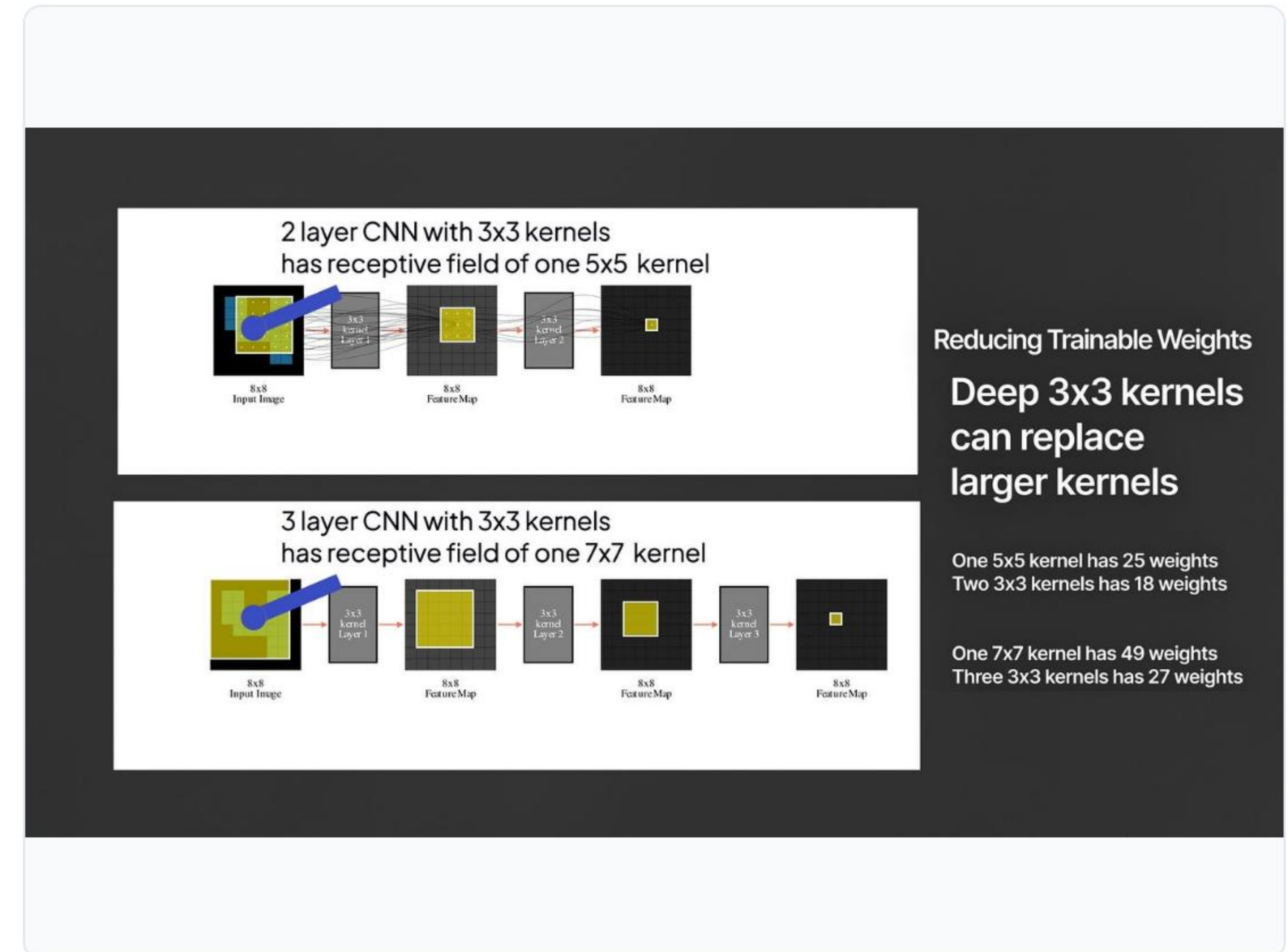
VGG-16 Architecture

- **Uniformity:** Exclusively uses **3x3 Convolutions** with stride 1 and padding 1.
- **Blocks:** Organized into 5 distinct convolutional blocks followed by max pooling.
- **Channel Growth:** Feature channels double after every pooling layer (64 → 128 → 256 → 512).
- **Depth:** Features 13 convolutional layers and 3 dense layers (16 weight layers total).



Why Multiple 3×3 Convolutions?

- 
Receptive Field Equivalence: Three stacked 3×3 layers cover the same spatial area as one 7×7 layer.
- 
Parameter Efficiency: Three 3×3 filters have 27 parameters ($3 * 9$). One 7×7 filter has 49 parameters.
~45% fewer parameters.
- 
More Non-Linearity: Stacking three layers means applying ReLU three times instead of once, allowing the network to learn more complex decision boundaries.

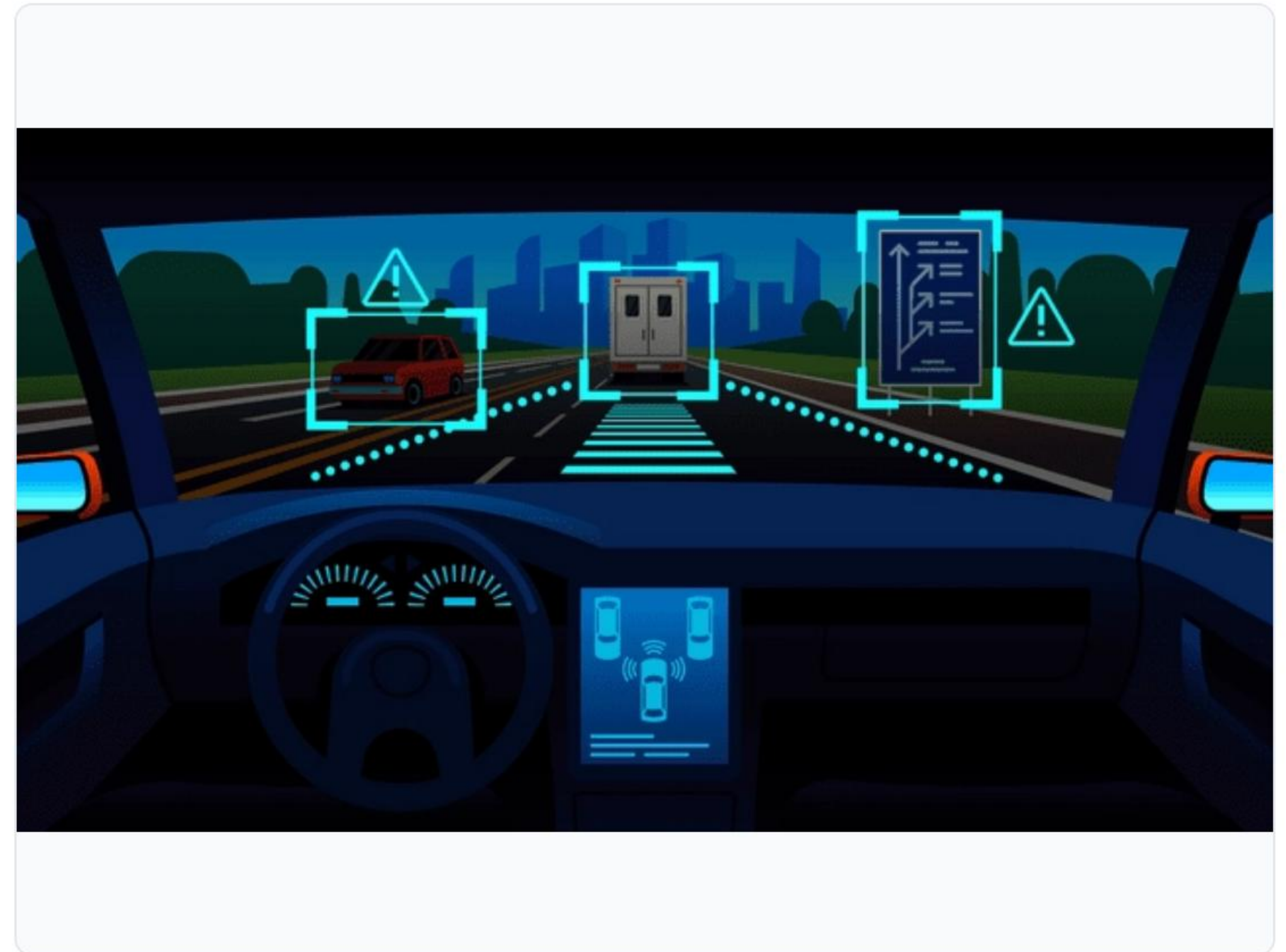


Architecture Summary Comparison

Metric	LeNet-5 (1998)	AlexNet (2012)	VGG-16 (2014)
Total Layers (Depth)	7	8	16
Parameter Count	~60 Thousand	~60 Million	~138 Million
Filter Sizes Used	5×5	11×11, 5×5, 3×3	Uniform 3×3
Key Architectural Innovation	Foundation of CNNs	ReLU, Dropout, GPUs	Deep uniform stacked blocks
Major Limitation	Cannot handle complex data	Chaotic architecture	Massive computational cost

Real-world Applications

-  **OCR Systems:** Modernized LeNet descendants are still used for reading documents and license plates.
-  **Face Recognition:** Deep architectures based on VGG principles extract highly detailed facial features.
-  **Medical Imaging:** VGG pre-trained networks are heavily used for tumor detection and X-ray analysis via transfer learning.
-  **Autonomous Vehicles:** Rapid object classification for pedestrian and vehicle detection relies heavily on CNNs.



Key Takeaways

- ✓ **Evolution via Hardware:** The jump from LeNet to AlexNet was primarily gated by the advent of GPU computing and large datasets like ImageNet.
- ✓ **Algorithmic Lifesavers:** Without **ReLU and Dropout**, training deep networks like AlexNet and VGG would have been mathematically impossible.
- ✓ **Simplicity over Complexity:** VGG proved that stacking simple, uniform layers (3×3) is superior to using varied, large convolutions.
- ✓ **Trade-offs:** Selecting an architecture requires balancing Accuracy vs. Parameter Cost (e.g., VGG is highly accurate but computationally heavy).

Class Discussion

Think-Pair-Share Activity:

If VGG-16 is so accurate, why don't we use it for every computer vision task today? Consider mobile phones and edge devices in your answer.

Preview of Next Lecture:

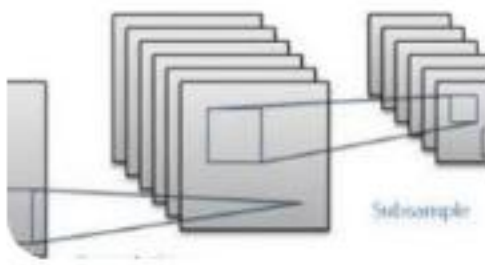
Landmark CNN Architectures II: ResNet and Inception

Image Sources



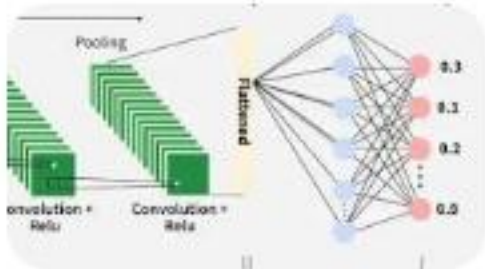
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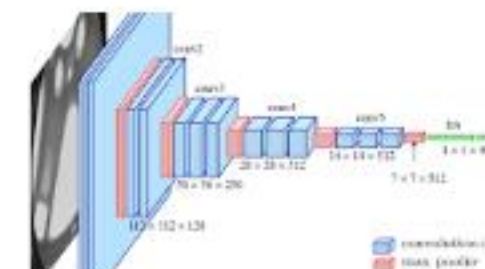
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Source: pyimagesearch.com



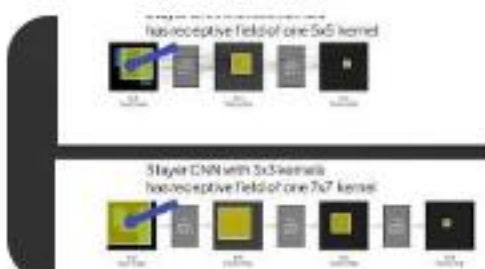
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